Computer Graphics Notes

Chapter 1 : Introduction

Computer Graphics

- Computer graphics is the science and art of communicating visually via a computer's display and its interaction devices. The visual aspect of the communication is usually in the computer-to-human direction, with the human-to-computer direction being mediated by devices like the mouse, keyboard, joystick, game controller, or touch-sensitive overlay.
- Computer graphics is a cross-disciplinary field in which physics, mathematics, human perception, human-computer interaction, engineering, graphic design, and art all play important roles.
- In the field of computer graphics, the word "model" can refer to a geometric model or a mathematical model. A **geometric model** is a model of something we plan to have appear in a picture: We make a model of a car, or a house, or an armadillo. The geometric model is enhanced with various other attributes that describe the color or texture or reflectance of the materials involved in the model. Starting from nothing and creating such a model is called **modeling**, and the geometric-plus-other-information description that is the result is called a **model.**
- A **mathematical** is a model of a physical or computational process.
- **Computer Graphics** are defined as any sketch or a drawing or a special network that pictorially represents some meaningful information.
- **Computer Graphics** is used where a set of images needs to be manipulated or the creation of the image in the form of pixels and is drawn on the computer.
- **Computer Graphics** can be used in digital photography, film, entertainment, electronic gadgets, and all other core technologies which are required. It is a vast subject and area in the field of computer science.
- **Computer Graphics** can be used in UI design, rendering, geometric objects, animation, and many more. In most areas, computer graphics is an abbreviation of **CG**.
- There are several tools used for the implementation of Computer Graphics. The basic is the <graphics.h> header file in Turbo-C, Unity for advanced and even OpenGL can be used for its Implementation.

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- It was invented in 1960 by great researchers Verne Hudson and William Fetter from Boeing.

Computer Graphics refers to several things:

- The manipulation and the representation of the image or the data in a graphical manner.
- Various technology is required for the creation and manipulation.
- Digital synthesis and its manipulation.

Types of Computer Graphics

- **Raster Graphics:** In raster, graphics pixels are used for an image to be drawn. It is also known as a bitmap image in which a sequence of images is into smaller pixels. Basically, a bitmap indicates a large number of pixels together.
- **Vector Graphics:** In vector graphics, mathematical formulae are used to draw different types of shapes, lines, objects, and so on.

Elements

The **seven** basic elements of graphic design are line, shape, color, texture, type, space and image.

1. **Lines** are always more than just points that are strung together. Depending on their form, weight, length and context, lines can help organize information, define shapes, imply movement, and convey emotions.

   When it comes to selecting the appropriate lines for projects, designers have plenty of options. **Lines** can:
   - …be **horizontal**, vertical or diagonal.
   - …be **straight**, curved or freeform.
   - …**zigzag** or create other patterns.
   - …be **solid**, broken or implied.
2. **Shapes** are best understood as areas, forms or figures contained by a boundary or closed outline. There are **two** types of shapes that every graphic designer should understand: **geometric** and **organic** (or "free-flowing").
   - **Geometric shapes** can include either two-dimensional or three-dimensional forms. Geometric shapes can include

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triangles, pyramids, squares, cubes, rectangles, pentagons, hexagons, octagons, decagons, circles, ellipses and spheres.

- **Organic shapes** are far less uniform, proportional and well-defined. They can be symmetrical or asymmetrical. They might include natural shapes, such as leaves, crystals, and vines, or abstract shapes, such as blobs and squiggles.

3. **Color** can be a useful tool for communicating a mood or provoking an emotional response from your viewer. Color theory and the color wheel provide a practical guide for graphic designers who want to select a single color or combine multiple colors in a harmonious- or intentionally discordant-way.
    - **Primary colors (red, yellow and blue)** are defined as the pure-pigment colors from which all others are made. There is no way to mix any other color to get red, yellow or blue. But mix them together, and you create all kinds of shades.
    - **Secondary colors (violet, green and orange)** are the immediate results of mixing two primary colors: Red and yellow make orange; blue and red make purple; and yellow and blue make green.
    - **Tertiary colors (red-orange, yellow-orange, yellow-green, blue-green, blue-violet and red-violet)** are the six colors that result from mixing a primary color and a secondary color.
4. **Texture** is the feel of a surface—furry, smooth, rough, soft, gooey or glossy. Most graphic designers must visually convey texture by using illusions to suggest how their work might feel if viewers could touch it.
5. **Type :** Whether you're choosing a font or creating your own typography for a graphic design project, it's important to make sure the type you use is legible and appropriate for your subject. Type affects the overall mood of a design, so consider whether your letters should be print or script, and whether they should have angles that are sharp or rounded.
6. **Space :** Spacing is a vital part of any designer's toolkit. It can give a design breathing room, increase its visual impact, balance out heavier visual elements, and emphasize images or messages that viewers should remember. Without enough space, a design can risk becoming too visually cluttered for your audience to understand.
7. **Image :** Whether graphic designers use photographs or illustrations, they rely on images to grab the audience's attention and express specific

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messages. An image works on multiple levels simultaneously: It provides context for a designer's communication, adds necessary drama or action, and creates an overall mood.

Applications

- **Computer Graphics are used for** an **aided design for engineering and architectural system-** These are used in electrical automobiles, electro-mechanical, mechanical, electronic devices. For example gears and bolts.
- **Computer Art –** MS Paint.
- **Presentation Graphics –** It is used to summarize financial statistical scientific or economic data. For example- Bar chart, Line chart.
- **Entertainment-** It is used in motion pictures, music videos, television gaming.
- **Education and training-** It is used to understand the operations of complex systems. It is also used for specialized system such for framing for captains, pilots and so on.
- **Visualization-** To study trends and patterns. For example- Analyzing satellite photo of earth.

Cathode Ray Tube (CRT):

CRT stands for Cathode Ray Tube. CRT is a technology used in traditional computer monitors and televisions. The image on CRT display is created by firing electrons from the back of the tube of phosphorus located towards the front of the screen.

Once the electron heats the phosphorus, they light up, and they are projected on a screen. The color you view on the screen is produced by a blend of red, blue and green light.

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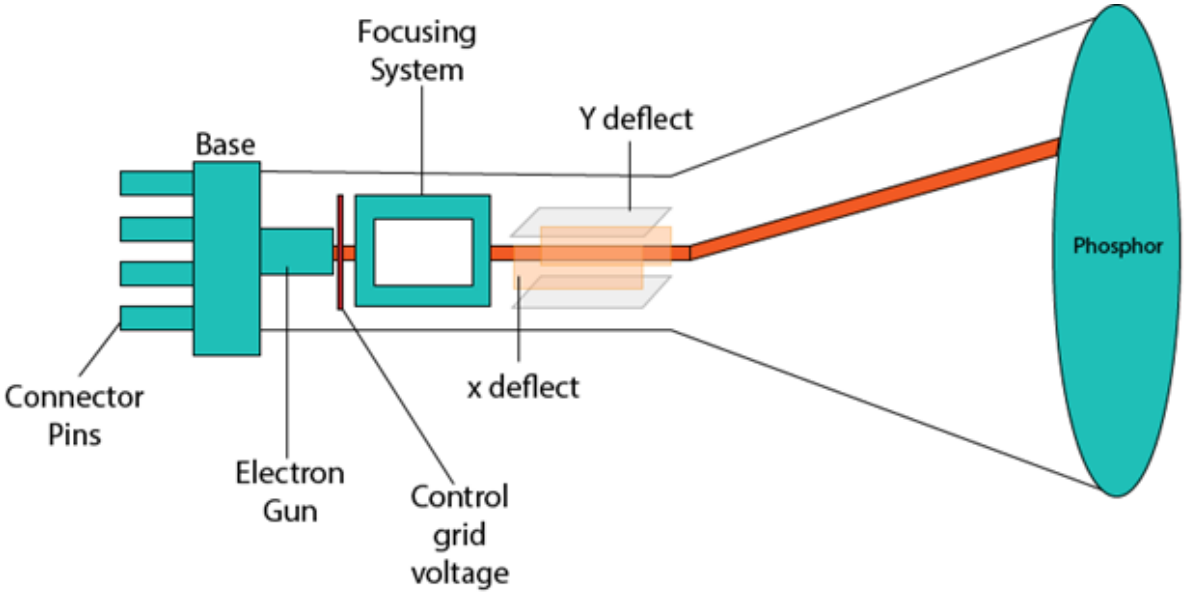

Cathode Ray Tube

Components of CRT:

Main Components of CRT are:

1. **Electron Gun:** Electron gun consisting of a series of elements, primarily a heating filament (heater) and a cathode. The electron gun creates a source of electrons which are focused into a narrow beam directed at the face of the CRT.

2. **Control Electrode:** It is used to turn the electron beam on and off.

3. **Focusing system:** It is used to create a clear picture by focusing the electrons into a narrow beam.

4. **Deflection Yoke:** It is used to control the direction of the electron beam. It creates an electric or magnetic field which will bend the electron beam as it passes through the area. In a conventional CRT, the yoke is linked to a sweep or scan generator. The deflection yoke which is connected to the sweep generator creates a fluctuating electric or magnetic potential.

5. **Phosphorus-coated screen:** The inside front surface of every CRT is coated with phosphors. Phosphors glow when a high-energy electron beam hits them. Phosphorescence is the term used to characterize the light given off by a phosphor after it has been exposed to an electron beam.

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Differentiate between Random and Raster Scan Display:

Random Scan	Raster Scan
1. It has high Resolution	1. Its resolution is low.
2. It is more expensive	2. It is less expensive
3. Any modification if needed is easy	3.Modification is tough
4. Solid pattern is tough to fill	4.Solid pattern is easy to fill
5. Refresh rate depends or resolution	5. Refresh rate does not depend on the picture.
6. Only screen with view on an area is displayed.	6. Whole screen is scanned.
7. Beam Penetration technology come under it.	7. Shadow mark technology came under this.
8. It does not use interlacing method.	8. It uses interlacing
9. It is restricted to line drawing applications	9. It is suitable for realistic display.

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